SENTHIL KUMAR, APPLIED AI ENGINEER

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- 🧢 👡 The hats I wear as an Applied AI Engineer:
  - An AWS Cloud Engineer: Architect & deploy backend AI systems in AWS
  - A Generative AI Engineer: Build agentic apps and LLM-powered products
  - An ML Engineer: Leverage self-hosted deep learning for inferencing in NLP applications
- ✍️ Passionate learner and technical writer: Writes at senthilkumarm1901.github.io and on Medium

WORK EXPERIENCE

Lead ML Engineer, Toyota Connected India **Jul 2022 – Present**

- Build production-grade LLM-workflows, RAGs and Agentic GenAI applications
- Design event-driven serverless systems on AWS (Lambda, API Gateway, DynamoDB, etc.).
- Lead a group of BE & FE engineers in AI & cloud projects
- Have published 4 articles in TCIN medium channel
- Have delivered a Talk on Serverless Python App in PyCon2023

Lead Data Scientist, Ford, Analytics Team **May 2018 – Jun 2022**

- A hands-on **data science developer** who co-developed with NLP experts from the US team
- Leveraged open source models for inferencing in dockerized NLP applications

Assistant Manager, LatentView Analytics (LV) **Apr 2014 – Apr 2018**

- Roles I played: **NLP Data Scientist**, **Project Delivery Manager**
- Responsible for project scoping and accountable for the delivery of Social Media Analytics projects of 8+ members

Lead Analyst, Beroe Inc **Jul 2010 – Dec 2013**

- Produce Market Research reports on how to procure indirect spend categories

PERFORMANCE AWARDS

- **Toyota** – Employee of the Month (Sep 2023); Hackathon Team Winner (Mar 2023); 2nd place, AWS India Manufacturing Teams Hackathon 2024; Winner, DriveDesign'25 & AWS Game Day Security Event '26
- **Ford** – Asia-Pacific Recognition Award (May 2019)
- **LatentView** – Encore Award (Jul–Sep 2016 Quarter)
- **Beroe** – Knowledge Contributor Award (twice, 2013)

EDUCATION

- Bachelors in Engg - Electronics - 8.6 CGPA | Madras Institute of Technology, 2006 - 2010
- State Topper in 2006 TN Engineering Entrance Exam (Physical Science 50/50); 12th Grade 95%, 10th Grade 92%

TECHNICAL UPSKILLING

- A 3-Course Specialization on AWS GenAI & Bedrock, Coursera-AWS, Jan'26
- Modern AI Agents, O'Reilly, Aug'25 | Building Apps with Anthropic's MCP, DeepLearning.AI, Jun'25
- A Proctored Exam on Git, Linux Foundation, Jan'25
- DevOps on AWS Specialization(4 courses), Coursera-AWS, Oct'24
- Generative AI with LLMs, Coursera-Deeplearning.ai, Nov'23
- DeepLearning Specialization(5 courses), Coursera-Deeplearning.ai, Dec'18 - May'19

Skills

- **Languages**: Python, SQL(basics), Markdown, Bash/Shell Scripting
- **Python Libraries**: Boto3, LangGraph, Strands, FastAPI, PyTorch, Transformers, SpaCy,
- **Tools**: Git, AWS CLI/Serverless/Bedrock, Docker, Python Env Management, Agentic Enginering

- 👀 **Curious what have I been cooking lately?** 🔗 LinkedIn Featured Section

KEY PROJECTS

Learn by Blogging | 35+ Articles | 2020 - Present

- Have been passionately writing about coding, GenAI, Cloud and ML since 2020 here and in Medium

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Agentic RAG Application | 2025 **More Details**

- Developed Agentic RAG application using Claude Opus model with
  - AWS Bedrock Agent powering the planning, reasoning, and reflection loops, and conversation memory, and
  - AWS OpenSearch for vector database knowledgebase, and semantic retrieval
- Extended the RAG architecture with MCP-tool execution to enable secure, production-grade tool-augmented inference | More Details

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GenAI Platform (X) with a Suite of Applications in AWS | 2024-25

- Led the development of a unified platform to derive actionable insights from text data
- Key products built via LLM-powered Pipelines:
  - xWorkFlow - Workflow automation tool with a YAML-based prompt engineering system for LLM chaining.
  - xAgenticChat - Secure answer engine for customer data interaction
  - xAnalyticsDashboard – Call center dashboard showcasing metrics from agent-customer interactions

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A Restricted Template Chatbot System powered by GenAI | 2025-26

- Built a restricted, template-based chatbot system enables reliability, control, and explainability, the key attributes missing in today's generative chatbots.
- Developed a 7-node workflow prototype LangGraph AI agent powered by state machine and template-based responses

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Serverless LLM-powered Classification Application | 2023

- Leveraged OpenAI paid models (such as GPT-4) to perform hierarchical text classification
- Utilised AWS Lambda functions to throttle the OpenAI API calls at scale

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Contact Center Dashboard - A Speech2Text Dashboard Application in AWS | 2022-23

- Architected a Speech-to-Text dashboard in AWS
- Powered by open-source ML models for speaker diarization and speech-to-text processing

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BERT Fine-tuned Aspect-based Sentiment Analysis Pipeline | 2020-21 **More Details**

- Built a ML Classification Pipeline to convert customer comments into Aspect and Sentiment pairs

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Personally Identifiable Information (PII) Detection using Named Entity Recognition (NER) | 2020-21 **More Details**

- Annonymized PII in text data by building a NER system using **RoBerta Fine-tuned Transformer model**
